

Nurein Umeya

nurein.umeya@alumni.utoronto.ca

GitHub: github.com/cl0udsw1tch

PROFESSIONAL SUMMARY

Software engineer and strong with algorithms, mainly those in general computer science, path planning, machine learning, and state estimation. Also advanced in pure mathematics, and great with low-level production environments interacting with the kernel and memory directly. My primary personal hobby is developing an archive of technical focus at <https://thecontraction.vercel.app>.

TECHNICAL SKILLS

Use purely UNIX environments, or anything conforming to the POSIX standard.

- **Programming:** C++, Python, Java, Golang, TypeScript, Bash, SQL
- **Machine Learning & AI:** PyTorch, TensorFlow
- **Cloud & DevOps:** AWS, GitLab CI/CD
- **Robotics & Embedded Systems:** ROS, SLAM, State Estimation, Nvidia CUDA
- **Web & Software Development:** React, Node.js, Express.js, Next.js, Flask, Svelte

EXPERIENCE

ML Engineer Intern | RBC | May 2025– May 2026

- Built cutting edge ML solutions throughout the research and product development lifecycle
- Played a key role in the design and development of Borealis' machine learning products
- To partner with RBC Borealis's research and product teams to ensure the seamless delivery of these products
- Applied engineering and data best practices to build robust and scalable large-scale machine learning software systems
- Supported projects with thorough documentation, design decisions, and technical advisory.

Software Engineer Intern | Github | May 2024 – May 2025

- Designed, built and deployed agentic solutions that leverage large language models to detect and prevent fraud, abuse, and security threats at scale - applying LLMs to problems such as content classification and multi-step agentic investigation.
- Built well-engineered, production-grade systems that run reliably against high-volume event streams, making effective use of AI coding assistants to accelerate and improve your work.
- Built and operated scalable ML systems on cloud platforms (such as Azure AI Foundry) for training, deploying, and serving models and agentic solutions in production.
- Evaluated and improved existing models and agentic solutions using offline evaluations (including tool-use loops and LLM-as-judge evaluation), performance metrics, and feedback from operational deployments.
- Documented the systems to help build and support the technical growth..

Software Engineer Intern | LeddarTech | Sept 2023 – May 2024

- Developed real-time road lane detection for motor vehicles using C++ and computer vision techniques.
- Optimized performance to achieve 15 FPS on embedded Nvidia devices.
- Utilized CI/CD tooling (Bitbucket, Git, Docker) in a Linux development environment.
- Performed extensive code reviews and debugging using GDB.

Teaching Assistant – Mathematics | University of Toronto | Sept 2021 – May 2022

- Led tutorials and graded assignments for Linear Algebra, Multivariable Calculus, and Vector Calculus courses.
- Provided tutoring and mentorship to students outside of lecture hours.

EDUCATION

Master of Engineering (Mechanical – Robotics & ML/Data Science) | *University of Toronto* | Jan 2023 – July 2025

- Relevant Coursework: Path Planning, Perception, Machine Learning, Big Data Analytics, State Estimation

Bachelor of Science (Computer Science, Mathematics) | *University of Toronto* | Sept 2017 – Apr 2022